

Figure captions

Fig. 1 FE-SEM images of a) high aspect ratio CNT b) low aspect ratio CNT and c) UHMWPE (PE) powder. Figure (d and e) shows the HAR and LAR CNT dispersion on PE powder, respectively, before sintering.

Fig. 2 DSC curve recorded with PE powder

Fig. 3 Fracture surface of 0.1 wt. % HARC (a) and LARC (b) composite

Fig. 4 DSC analysis for different CNT reinforced PE and pure PE composite.

Fig. 5 Thermo gravimetric curves of composite revealing degradation loss through temperature range up to 500° C

Fig. 6 The representative load displacement curves obtained from nanoindentation tests for prepared composite

Fig. 7 Hardness and Elastic modulus of different compositions obtained from nanoindentation

Fig. 8 Graphical representation of the effect of CNT aspect ratio while interacting with polymer chains

Fig. 9 Regression analysis for the experimental and theoretical elastic modulus of the HARC composite (a) and LARC composite (b)

Fig. 10 Variation of friction co-efficient over the sliding distance for the different compositions

Fig. 11 Wear track surface profile for the prepared composites

Fig. 12 variation of Specific wear rate for prepared composites

Fig. 13 FE-SEM images of worn surface showing debris contained rough surface in pure PE (a and a') and smooth features in PE-0.1LARC (b-b') and PE-0.1HARC (c-c')







